Virtual Private Networking

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In the cyber-village world of today, VPN is a safe haven for chosen connectivity

In the digital age, it is not necessary for a team of workers to be in the same building. They can collaborate without even meeting each other, forming a 'virtual team'. Given the current connectivity scenario, a security consultant in Bangalore can work with a system administrator in New York, using inputs from a colleague in Berlin.

When companies expand operations, the biggest problem they face is that of connecting to their branch offices. A leased line is a solution if the company has only a few offices; when the number of offices increases, the number of connections required to inter-connect them increases exponentially, along with cost. This is where Virtual Private Networking (VPN) comes in.

VPN, as the name indicates, is a private network used only by a restricted set of people, and has no actual physical connections between nodes. The network is built by establishing virtual connections via a public network, such as the Internet. Since data is transmitted over the Internet, encryption and security play a vital role in VPN.

Generally, VPNs can be used in two ways—the first is Remote Access, wherein employees can access the company's Intranet from remote locations, such as their homes. The second is to connect various offices of a company, also called site-to-site connections. This technology does not offer anything unique per se, and can be considered an evolution of the Local Area Network (LAN) and Wide Area Netowork (WAN) technologies. A combination of these two is also possible.

The ingredients

Data is sent between two networks as discrete units called packets. The format of the packet varies depending on the protocols used. When you try to connect two different networks, you may have to send a packet that the destination network cannot understand. Tunnelling provides a simple solution to this problem by encapsulating a packet within another packet, which the destination network will be able to understand. VPN uses this technique to connect different networks.

Generally, a VPN consists of two components—the gateway and the client. The gateway is a combination of hardware and software, set up at the company's physical location. Each LAN has its own VPN gateway, which is the entry and exit point of the LAN. A VPN gateway can communicate with more than one client simultaneously, by using multiple